$a_n = f_{n-1}$

$b_n = a_{n-1}$

Hw one 7.1 p. 458 Q24:

a) f_n = # of n length strings that have AAA. $\rightarrow$ [A, B]

s_n = # of n length strings that dont have AAA. $\rightarrow$ {A, B}

$\rightarrow f_n = 2^n - s_n$

$a_n = f_n^0$: # of strings that end with zero consecutive A's

$b_n = f_n^1$: # of strings that end with one A.

$c_n = f_n^2$: # of strings that end with two A's.

$f_n = a_n + b_n + c_n$

Extending string length of f_n $(n+1)$:

$a_n \rightarrow$ can either add A or B to the end.

$\hookrightarrow$ Adding B: a_{n+1} (Ends with B)

$\rightarrow$ ~~Adding A: Becomes b_{n+1}~~ (Ends with one A)

$b_n \rightarrow$ Ends with one A

$\hookrightarrow$ Add B : a_{n+1}

$\hookrightarrow$ Add A : Becomes c_{n+1} (Ends with two A's now)

$c_n \rightarrow$ Ends with two A's

$\hookrightarrow$ Adding B: a_{n+1}

$\hookrightarrow$ ~~Adding A~~: Becomes string with three A's, not allowed, can't add A.

$a_{n+1} = \underbrace{a_n \text{ (Adding B)}}_{\hookrightarrow B} + \underbrace{b_n \text{ (Adding B)}}_{\hookrightarrow BA} + \underbrace{c_n \text{ (Adding B)}}_{\hookrightarrow BAA}$

$\hookrightarrow = a_n + b_n + c_n = f_n$

$b_{n+1} = a_n \cdot \text{(adding A)}] \rightarrow BA$

$c_{n+1} = b_n \cdot \text{(adding A)}] \rightarrow AA$

Using Simple Conversions!

$f_{n+1} = a_{n+1} + b_{n+1} + c_{n+1} = f_n + a_n + b_n$

$f_n = a_n + b_n + c_n = f_{n-1} + a_{n-1} + a_{n-2} = f_{n-1} + f_{n-2} + f_{n-3}$

$\hookrightarrow f_n = f_{n-1} + f_{n-2} + f_{n-3}$

$\hookrightarrow$ Recurrence for strings that dont contain AAA

$a_n = f_{n-1}$ $\quad f_n = a_{n+1}$

$b_n = a_{n-1}$ $\quad a_n = b_{n+1}$

$c_n = b_{n-1} = a_{n-2}$

$a_{n-1} = f_{n-2}$ $\quad a_{n-2} = f_{n-3}$

$c_{n+1} = b_n$